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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Decisive
Selected Domain	Domain 1 — The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth (Primary) | Secondary: SDG 4, 9, 10, 13
Project / Solution Name	Nexus (NexusTalent — A Reviewable Reskilling Decision Workspace)
Name (Leader)	Raye Chan Jun Foong
Email & Phone (Leader)	rayechanjf@gmail.com 0198184052
University Affiliation(s)	University Malaya
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI advancement and industrial transitions are changing job tasks faster than traditional training programmes and static skills frameworks can be updated. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025), while the Ministry of Science, Technology and Innovation estimates over 600,000 workers will need reskilling within three to five years.
This creates a critical decision bottleneck for three key groups:

1. **Working Adults & Career Switchers:** Workers like operations coordinators with finite study budgets (e.g., 4 hours/week) do not know which existing task strengths transfer, which skill gaps deserve their next learning hour, or what evidence justifies changing direction. Static pathway lists and black-box chatbots fail to show source provenance, ignore prerequisite ordering, and erase completed learning progress whenever market scenarios shift.
2. **TVET & University Career Advisers:** Advisers lack reviewable, evidence-backed transition briefs to inspect why a recommendation changed, verify dated sources, or revise a plan without losing prior work.
3. **HR & Employers:** Internal mobility and L&D teams lack an explainable way to survey department skills, visualize team readiness via heatmaps/radars, compare AI or green-transition scenarios across cohorts, and assemble cited reskilling cohorts without exposing sensitive personal data.
Socially and economically, unguided reskilling wastes scarce study hours. Workers, advisers, and employers need a transparent decision workspace where plans evolve as technology moves, while completed progress stays preserved.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Nexus (NexusTalent) is a reviewable reskilling decision workspace for workers, career advisers, and HR/employers deployed live on Replit (https://NexusTalent-Decisive.replit.app). Rather than acting as an opaque chatbot, Nexus provides an individual six-stage worker loop (*Review experience, Compare directions, Inspect evidence, Build a plan, Review a change, Retain learning*) alongside a seven-chapter **HR & Employers workflow (/hr/demo)** . Our solution delivers all six Domain 1 must-have features through One shared explainable rule and signal engine :

1. **Skills Profile & Cohort Intake:** Workers upload a PDF/DOCX/TXT CV locally in the browser or load fictional profiles (e.g., Siti Rosli). For employers, HR manages three seeded departments (24 synthetic profiles) and generates expiring, quota-limited /survey/:token links where employees submit profiles and immediately receive explainable role matches .
2. **Updatable AI-Trend Signal Layer:** An Express service retrieves live public research metadata via the Crossref REST API (recording DOI, publication/retrieval dates, and cache status) and applies transparent keyword-to-skill mapping rules .
3. **Explainable Future-Role Matching:** Six authored roles are ranked using a 100-point Match Index (45% required skills, 30% transferable task bridges, 25% research/scenario context), surfaced individually or across department radars, heatmaps, and searchable employee reviews .

4. **Personalised & Cohort Pathways:** Generates prerequisite-ordered learning steps scheduled to weekly hour budgets using free public resources, plus selectable HR learning cohorts with downloadable cited briefs .
5. **UN SDG Mapping:** Links every role, skill, and learning cohort to specific UN SDG targets (SDG 4.4, 8.2, 8.5, 9.5, 10.2, 12.6) .
6. **Handling Uncertainty & Retaining Progress:** Both workers and HR compare *Steady adoption*, *Faster AI adoption*, and *Green transition* scenarios (plus HR trend simulations), inspecting diffs while preserving completed learning via stable step IDs (skill:dataQuality) .

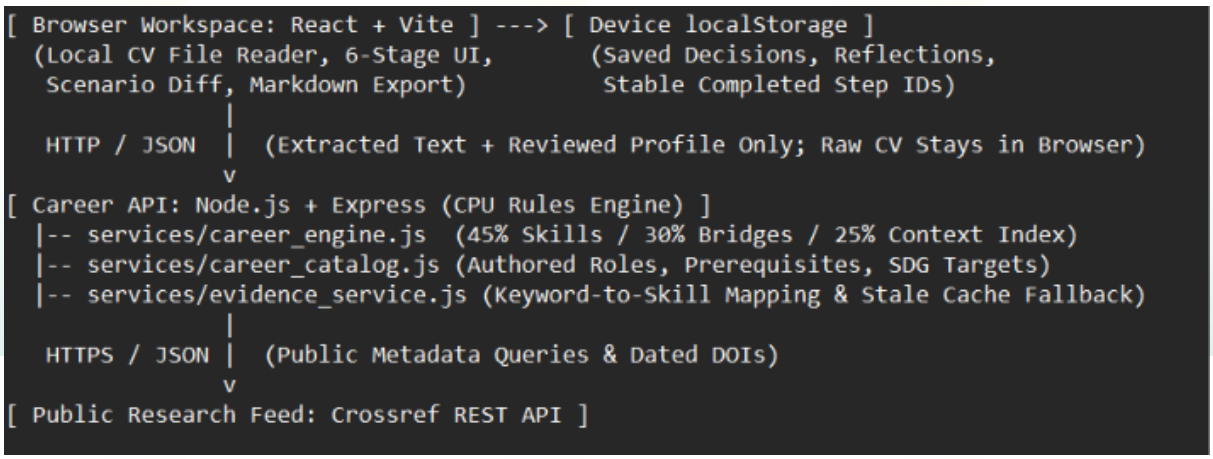
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Frontend & Presentation (src/): React 18 and Vite SPA styled with custom CSS design tokens (src/career.css, src/landing.css) and embedded Sora, DM Sans, and IBM Plex Mono typography, hosted live on Replit (https://NexusTalent-Decisive.replit.app) . Handles local CV file reading, interactive 6-stage reskilling workspace, scenario diff review, and Markdown career brief export.**
- **Backend API & Rules Engine (services/): Node.js with Express (services/career_engine.js, services/evidence_service.js) hosted on Replit, running deterministic CPU scoring rules, prerequisite pathway ordering, and public research signal caching over RESTful HTTPS/JSON endpoints.**
- **Knowledge, Catalog & Persistence Layer: Authored role, skill, scenario, and public learning resource catalogs (src/data/careerCatalog.js, services/career_catalog.js), a server-side versioned JSON revision cache for retrieved Crossref metadata, and browser localStorage for device-local persistence of user decisions, reflections, and completed learning step IDs.**
- **External Signal Feed: Public Crossref REST API (https://api.crossref.org/works) for dated scholarly metadata and DOIs on AI and workforce trends.**
- **Official SDC Sponsor Resources Used:**
- **Replit: Used to build, host, and deploy the live full-stack React/Vite frontend and Node.js/Express Career API web application (https://NexusTalent-Decisive.replit.app) .**
- **System Block Diagram:**



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Through design-thinking, we framed two connected moments of need: a working adult (like Siti, an operations coordinator with 4 hours/week to study) reviewing her next learning hour with an adviser, and an HR team planning department-wide reskilling cohorts . **Empathize & Define:** Workers distrust black-box AI scores that hide assumptions and erase progress when plans change. Advisers and HR managers need auditable transition briefs, department skill heatmaps, and cohort plans—not unverified course lists . **Ideate & Prototype:** We designed a six-stage individual workspace and a seven-chapter HR & Employers demo (/hr/demo) powered by the same deterministic matching engine. HR can launch expiring, quota-limited /survey/:token campaigns (updating every 5 seconds), inspect team radars/heatmaps, simulate scenarios, and export cited cohort briefs . **Test & Deploy:** We verified end-to-end continuity using Node unit tests and Playwright browser integration tests across intake, scenario accept/revert, progress preservation (skill:dataQuality), and HR survey flows. Hosted live on Replit (https://NexusTalent-Decisive.replit.app) with zero GPU/LLM dependencies, individual CVs stay in browser localStorage while HR demo campaigns use synthetic-only server JSON storage

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: Target 4.4 ("By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship") and Target 8.2 ("Achieve higher levels of economic productivity through diversification, technological upgrading and innovation"). Secondary: Target 8.5 (full and productive employment), Target 9.5 (enhance scientific research and technical capabilities), Target 10.2 (inclusive participation), and Target 12.6 (sustainable corporate practices and reporting).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In Malaysia, while 56% of professionals already use AI tools at work, only 26% have received formal training (Hays Asia Salary Guide 2025), and over 600,000 workers require reskilling within 3–5 years (MOSTI)—yet static learning lists provide 0% continuity of recorded progress when career plans are revised.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Time to Adviser-Agreed Next Action (minutes from profile intake to an exported, adviser-reviewed transition brief). 2. Source Provenance & Link Validity Rate (% of recommended roles and signals backed by inspectable, dated DOI/URL receipts). 3. Plan Revision Continuity Rate (% of completed learning

Metric Category	Your Response
	steps preserved across accepted or reverted scenario revisions). 4. Completed Practice Work-Sample Review Rate (adviser verification of learner work samples).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. Enable a worker and adviser to reach a documented, source-cited next learning step in **<10 minutes** across a 2-session pilot. 2. **100% source provenance transparency** (every signal cites publisher, publication date, retrieval timestamp, and mapping rule) and **100% retention of completed learning steps** across scenario switches. 3. Validate transition brief usefulness and work-sample completion with **1 career adviser team across 2 review sessions** (pilot target; baseline pending).

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React & React DOM	Code Framework / Library	MIT License
Vite	Build Tool / Dev Server	MIT License
Node.js & Express	Backend Runtime & Web Framework	MIT License
Crossref REST API Metadata	Public Scholarly Metadata Feed	Public Facts / CC0 1.0 (Crossref Metadata)
Public Course Links (Microsoft Learn, UK Gov Data Quality Framework)	External Public Learning Resources (Catalogued URLs)	Publicly Accessible Web Documentation
Sora, DM Sans, IBM Plex Mono	Open-Source Typography	SIL Open Font License 1.1

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

23/10/26